



SQL Data Types for MySQL, SQL Server, and MS Access

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The data type of a column defines what value the column can hold: integer, character, money, date and time, binary, and so on.

SQL Data Types

Each column in a database table is required to have a name and a data type.

An SQL developer must decide what type of data that will be stored inside each column when creating a table. The data type is a guideline for SQL to understand what type of data is expected inside of each column, and it also identifies how SQL will interact with the stored data.

Note: Data types might have different names in different database. And even if the name is the same, the size and other details may be different! **Always check the documentation!**

MySQL Data Types (Version 8.0)

In MySQL there are three main data types: string, numeric, and date and time.

String Data Types

Data type	Description
CHAR(size)	A FIXED length string (can contain letters, numbers, and special characters). The <i>size</i> parameter specifies the column length in characters - can be from 0 to 255. Default is 1
VARCHAR(size)	A VARIABLE length string (can contain letters, numbers, and special characters). The <i>size</i> parameter specifies the maximum string length in characters - can be from 0 to 65535
BINARY(size)	Equal to CHAR(), but stores binary byte strings. The <i>size</i> parameter specifies the column length in bytes. Default is 1
VARBINARY(size)	Equal to VARCHAR(), but stores binary byte strings. The <i>size</i> parameter specifies the maximum column length in bytes.
TINYBLOB	For BLOBs (Binary Large Objects). Max length: 255 bytes
TINYTEXT	Holds a string with a maximum length of 255 characters
TEXT(size)	Holds a string with a maximum length of 65,535 bytes
BLOB(size)	For BLOBs (Binary Large Objects). Holds up to 65,535 bytes of data
MEDIUMTEXT	Holds a string with a maximum length of 16,777,215 characters
MEDIUMBLOB	For BLOBs (Binary Large Objects). Holds up to 16,777,215 bytes of data
LONGTEXT	Holds a string with a maximum length of 4,294,967,295 characters
LOBLOB	For BLOBs (Binary Large Objects). Holds up to 4,294,967,295 bytes of data

ENUM(val1, val2, val3, ...)

A string object that can have only one value, chosen from a list of possible values. You can list up to 65535 values in an ENUM list. If a value is inserted that is not in the list, a blank value will be inserted. The values are sorted in the order you enter them

SET(val1, val2, val3, ...)

A string object that can have 0 or more values, chosen from a list of possible values. You can list up to 64 values in a SET list

Numeric Data Types

Data type	Description
BIT(<i>size</i>)	A bit-value type. The number of bits per value is specified in <i>size</i> . The <i>size</i> parameter can hold a value from 1 to 64. The default value for <i>size</i> is 1.
TINYINT(<i>size</i>)	A very small integer. Signed range is from -128 to 127. Unsigned range is from 0 to 255. The <i>size</i> parameter specifies the maximum display width (which is 255)
BOOL	Zero is considered as false, nonzero values are considered as true.
BOOLEAN	Equal to BOOL
SMALLINT(<i>size</i>)	A small integer. Signed range is from -32768 to 32767. Unsigned range is from 0 to 65535. The <i>size</i> parameter specifies the maximum display width (which is 255)
MEDIUMINT(<i>size</i>)	A medium integer. Signed range is from -8388608 to 8388607. Unsigned range is from 0 to 16777215. The <i>size</i> parameter specifies the maximum display width (which is 255)
INT(<i>size</i>)	A medium integer. Signed range is from -2147483648 to 2147483647. Unsigned range is from 0 to 4294967295. The <i>size</i> parameter specifies the maximum display width (which is 255)
INTEGER(<i>size</i>)	Equal to INT(<i>size</i>)
BIGINT(<i>size</i>)	A large integer. Signed range is from -9223372036854775808 to 9223372036854775807.

	Unsigned range is from 0 to 18446744073709551615. The <i>size</i> parameter specifies the maximum display width (which is 255)
FLOAT(<i>size</i> , <i>d</i>)	A floating point number. The total number of digits is specified in <i>size</i> . The number of digits after the decimal point is specified in the <i>d</i> parameter. This syntax is deprecated in MySQL 8.0.17, and it will be removed in future MySQL versions
FLOAT(<i>p</i>)	A floating point number. MySQL uses the <i>p</i> value to determine whether to use FLOAT or DOUBLE for the resulting data type. If <i>p</i> is from 0 to 24, the data type becomes FLOAT(). If <i>p</i> is from 25 to 53, the data type becomes DOUBLE()
DOUBLE(<i>size</i> , <i>d</i>)	A normal-size floating point number. The total number of digits is specified in <i>size</i> . The number of digits after the decimal point is specified in the <i>d</i> parameter
DOUBLE PRECISION(<i>size</i> , <i>d</i>)	
DECIMAL(<i>size</i> , <i>d</i>)	An exact fixed-point number. The total number of digits is specified in <i>size</i> . The number of digits after the decimal point is specified in the <i>d</i> parameter. The maximum number for <i>size</i> is 65. The maximum number for <i>d</i> is 30. The default value for <i>size</i> is 10. The default value for <i>d</i> is 0.
DEC(<i>size</i> , <i>d</i>)	Equal to DECIMAL(<i>size</i> , <i>d</i>)

Note: All the numeric data types may have an extra option: UNSIGNED or ZEROFILL. If you add the UNSIGNED option, MySQL disallows negative values for the column. If you add the ZEROFILL option, MySQL automatically also adds the UNSIGNED attribute to the column.

Date and Time Data Types

Data type	Description
DATE	A date. Format: YYYY-MM-DD. The supported range is from '1000-01-01' to '9999-12-31'
DATETIME(<i>fsp</i>)	A date and time combination. Format: YYYY-MM-DD hh:mm:ss. The supported range is from '1000-01-01

	00:00:00' to '9999-12-31 23:59:59'. Adding DEFAULT and ON UPDATE in the column definition to get automatic initialization and updating to the current date and time
TIMESTAMP(<i>fsp</i>)	A timestamp. TIMESTAMP values are stored as the number of seconds since the Unix epoch ('1970-01-01 00:00:00' UTC). Format: YYYY-MM-DD hh:mm:ss. The supported range is from '1970-01-01 00:00:01' UTC to '2038-01-09 03:14:07' UTC. Automatic initialization and updating to the current date and time can be specified using DEFAULT CURRENT_TIMESTAMP and ON UPDATE CURRENT_TIMESTAMP in the column definition
TIME(<i>fsp</i>)	A time. Format: hh:mm:ss. The supported range is from '-838:59:59' to '838:59:59'
YEAR	A year in four-digit format. Values allowed in four-digit format: 1901 to 2155, and 0000. MySQL 8.0 does not support year in two-digit format.

SQL Server Data Types

String Data Types

Data type	Description	Max size	Storage
char(n)	Fixed width character string	8,000 characters	Defined width
varchar(n)	Variable width character string	8,000 characters	2 bytes + number of chars
varchar(max)	Variable width character string	1,073,741,824 characters	2 bytes + number of chars
text	Variable width character string	2GB of text data	4 bytes + number of chars

nchar	Fixed width Unicode string	4,000 characters	Defined width x 2
nvarchar	Variable width Unicode string	4,000 characters	
nvarchar(max)	Variable width Unicode string	536,870,912 characters	
ntext	Variable width Unicode string	2GB of text data	
binary(n)	Fixed width binary string	8,000 bytes	
varbinary	Variable width binary string	8,000 bytes	
varbinary(max)	Variable width binary string	2GB	
image	Variable width binary string	2GB	

Numeric Data Types

Data type	Description	Storage
bit	Integer that can be 0, 1, or NULL	
tinyint	Allows whole numbers from 0 to 255	1 byte
smallint	Allows whole numbers between -32,768 and 32,767	2 bytes
int	Allows whole numbers between -2,147,483,648 and 2,147,483,647	4 bytes
bigint	Allows whole numbers between -9,223,372,036,854,775,808 and 9,223,372,036,854,775,807	8 bytes
decimal(p,s)	Fixed precision and scale numbers. Allows numbers from $-10^{38} + 1$ to $10^{38} - 1$. The p parameter indicates the maximum total number of digits that can be stored (both to the left and to the right of the decimal point). p must be a value from 1 to 38.	5-17 bytes

Default is 18.

The *s* parameter indicates the maximum number of digits stored to the right of the decimal point. *s* must be a value from 0 to *p*. Default value is 0

numeric(<i>p,s</i>)	Fixed precision and scale numbers. Allows numbers from $-10^{38} + 1$ to $10^{38} - 1$. The <i>p</i> parameter indicates the maximum total number of digits that can be stored (both to the left and to the right of the decimal point). <i>p</i> must be a value from 1 to 38. Default is 18. The <i>s</i> parameter indicates the maximum number of digits stored to the right of the decimal point. <i>s</i> must be a value from 0 to <i>p</i>. Default value is 0	5-17 bytes
smallmoney	Monetary data from -214,748.3648 to 214,748.3647	4 bytes
money	Monetary data from -922,337,203,685,477.5808 to 922,337,203,685,477.5807	8 bytes
float(<i>n</i>)	Floating precision number data from $-1.79E + 308$ to $1.79E + 308$. The <i>n</i> parameter indicates whether the field should hold 4 or 8 bytes. float(24) holds a 4-byte field and float(53) holds an 8-byte field. Default value of <i>n</i> is 53.	4 or 8 bytes
real	Floating precision number data from $-3.40E + 38$ to $3.40E + 38$	4 bytes

Date and Time Data Types

Data type	Description	Storage
datetime	From January 1, 1753 to December 31, 9999 with an accuracy of 3.33 milliseconds	8 bytes
datetime2	From January 1, 0001 to December 31, 9999 with an accuracy of 100 nanoseconds	6-8 bytes

smalldatetime	From January 1, 1900 to June 6, 2079 with an accuracy of 1 minute	4 bytes
date	Store a date only. From January 1, 0001 to December 31, 9999	3 bytes
time	Store a time only to an accuracy of 100 nanoseconds	3-5 bytes
datetimeoffset	The same as datetime2 with the addition of a time zone offset	8-10 bytes
timestamp	Stores a unique number that gets updated every time a row gets created or modified. The timestamp value is based upon an internal clock and does not correspond to real time. Each table may have only one timestamp variable	

Other Data Types

Data type	Description
sql_variant	Stores up to 8,000 bytes of data of various data types, except text, ntext, and timestamp
uniqueidentifier	Stores a globally unique identifier (GUID)
xml	Stores XML formatted data. Maximum 2GB
cursor	Stores a reference to a cursor used for database operations
table	Stores a result-set for later processing

MS Access Data Types

Data type	Description	Storage
Text	Use for text or combinations of text and numbers. 255 characters maximum	
Memo	Memo is used for larger amounts of text. Stores up to 65,536 characters. Note: You cannot sort a memo field. However, they are searchable	

Byte	Allows whole numbers from 0 to 255	1 byte
Integer	Allows whole numbers between -32,768 and 32,767	2 bytes
Long	Allows whole numbers between -2,147,483,648 and 2,147,483,647	4 bytes
Single	Single precision floating-point. Will handle most decimals	4 bytes
Double	Double precision floating-point. Will handle most decimals	8 bytes
Currency	Use for currency. Holds up to 15 digits of whole dollars, plus 4 decimal places. Tip: You can choose which country's currency to use	8 bytes
AutoNumber	AutoNumber fields automatically give each record its own number, usually starting at 1	4 bytes
Date/Time	Use for dates and times	8 bytes
Yes/No	A logical field can be displayed as Yes/No, True/False, or On/Off. In code, use the constants True and False (equivalent to -1 and 0). Note: Null values are not allowed in Yes/No fields	1 bit
Ole Object	Can store pictures, audio, video, or other BLOBs (Binary Large Objects)	up to 1GB
Hyperlink	Contain links to other files, including web pages	
Lookup Wizard	Let you type a list of options, which can then be chosen from a drop-down list	4 bytes

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